

Multiple Choice (10 marks)

1. A starvation-free job-scheduling policy guarantees that no job waits indefinitely for service. Which of the following job-scheduling policies is starvation-free?
 - (a) Round-robin
 - (b) Priority queuing
 - (c) Shortest job first
 - (d) Youngest job first
2. Which of the following decimal numbers has an exact representation in binary notation?
 - (a) 0.1
 - (b) 0.2
 - (c) 0.3
 - (d) 0.5
3. Which of the following is NOT a property of bitmap graphics?
 - (a) Fast hardware exists to move blocks of pixels efficiently.
 - (b) Realistic lighting and shading can be done.
 - (c) All line segments can be displayed as straight.
 - (d) Polygons can be filled with solid colors and textures.
4. Which of the following is usually NOT represented in a subroutine's activation record frame for a stack-based programming language?
 - (a) Values of local variables
 - (b) A heap area
 - (c) The return address
 - (d) Stack pointer for the calling activation record
5. Which of the following are most vulnerable to injection attacks?
 - (a) Session IDs
 - (b) Registry keys
 - (c) Network communications
 - (d) SQL queries based on user input

True / False (10 marks)

Answer True or False

6. True or False: Mutation-based fuzzing generates each different input by modifying a prior input to discover vulnerabilities.
7. True or False: The AH Protocol guarantees both data integrity and nonrepudiation while providing limited privacy features.
8. True or False: Message integrity ensures that data arrives at the receiver exactly as it was sent, without alterations during transmission.
9. True or False: When the STP solver times out on a constraint query, EXE assumes the query is not satisfiable and stops executing the path.
10. True or False: Busy-waiting can be justified if the processor has no other tasks to perform during the wait period.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the slope of a line.
12. Write a function to convert snake case string to camel case string.

End of Paper